

Question 1.**Marks**

- (a) Given $z_1 = 2 - 3i$ and $z_2 = 3 + 4i$,
- (i) Find: $|z_1|$. 1
- (ii) Find: $z_1 + \bar{z}_2$. 1
- (iii) Find: $\frac{z_2}{z_1}$. 2
- (b) If $(a + 3i)(7 - i) = 17 + bi$, where a and b are real numbers. Find the value of b . 2
- (c) Given $\Omega^2 = 35 - 12i$, find Ω . 2
- (d) Show that: $(1 + i)^{2011} = 2^{1005}(-1 + i)$. 3
- (e) Suppose $\alpha = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$ and $\beta = 1 - i$,
- (i) Find $\arg\beta$. 1
- (ii) Find the smallest positive integer m such that $\frac{\alpha^m}{\beta^m}$ is purely imaginary. 2
- (iii) For this value of m , find the value of $\frac{\alpha^m}{\beta^m}$. 1

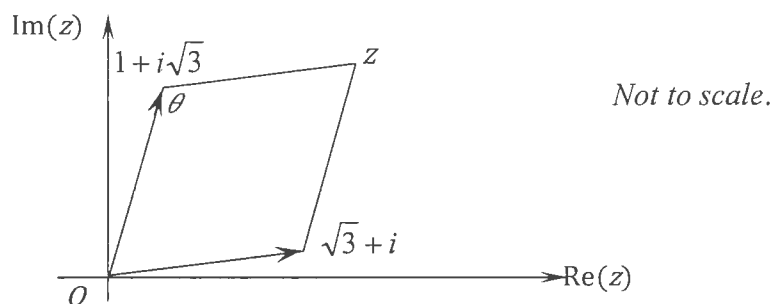
Question 2. [Start a new page]

- (a) Find the complex number $a + ib$ when $1 + 2i$ is rotated about the point $(3, 1)$ by $\frac{\pi}{2}$ in an Argand plane. 2
- (b) Shade the region on an Argand plane satisfying z for $\left|\frac{1}{z} + 1\right| \geq 1$. 1
- (c) It is known that $5 - 6i$ is a zero of the polynomial function:
 $P(z) = 2z^3 - 19z^2 + 112z + a_0$, where a_0 is real.
- (i) Find the other two zeros of $P(z)$. 2
- (ii) Find the value of a_0 . 2
- (d) On an Argand plane, sketch the region described for z when:
 $\left|\frac{1}{z} + \frac{1}{\bar{z}}\right| \geq \frac{1}{2}$ and $\text{Im}(z) \geq 0$ and $\frac{\pi}{6} \leq \arg z \leq \frac{3\pi}{4}$. 4
- (e) Given that ω is one of the complex cube roots of unity,
- (i) Show that ω^2 can be the other complex root. 1
- (ii) What is the value of $1 + \omega + \omega^2$. 1
- (iii) Find: $(\omega - 1)(1 + 2\omega + 3\omega^2)$ 2

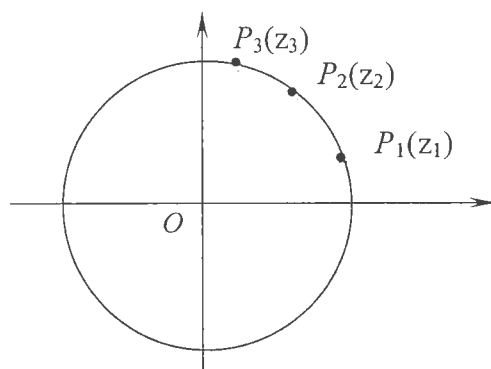
Question 3. [Start a new page]

Marks

- (a) Given $z = \cos \theta + i \sin \theta$, where $z \neq 0$.
- (i) Show that: $\cos(n\theta) = \frac{z^n + z^{-n}}{2}$. 2
- (ii) Hence show that $\cos \theta \cdot \cos 2\theta = \frac{1}{2}(\cos \theta + \cos 3\theta)$. 2
- (b) On the Argand diagram, the complex numbers 0 , $\sqrt{3} + i$, z and $1 + i\sqrt{3}$ form a rhombus as shown.



- (i) Find z in the form $a + ib$, where a and b are real numbers. 1
- (ii) Find the value of θ , the marked interior angle of the rhombus. 2
- (c) Find the locus of z when: $\arg(z + 2) + \arg(z - 2) = \pi$. 3
- (d) For the three complex numbers z_1, z_2 and z_3 , 3
- If $|z_1| = |z_2| = |z_3|$ such that $0 < \arg z_1 < \arg z_2 < \arg z_3 < \frac{\pi}{2}$,
as indicated in the diagram



On the diagram provided (see last page), explain why $\arg\left(\frac{z_3 - z_2}{z_3 - z_1}\right) = \frac{1}{2} \arg\left(\frac{z_2}{z_1}\right)$.

Question 3 continued over the page:

Question 3 continued

- (e) Which one of the following Argand planes below could represent the position of the roots of $z^5 + z^2 - z + k = 0$, where k is a real number. Give reasons. 2

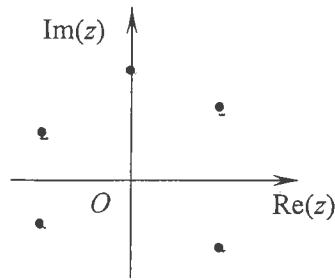


Diagram A

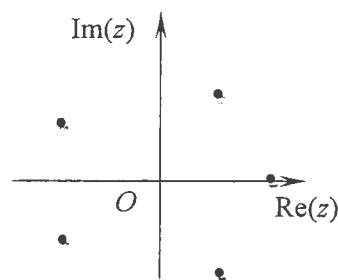


Diagram B

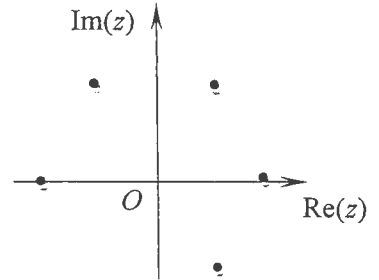


Diagram C

Question 4. [Start a new page] Marks

- (a) Find the locus of z when: $\frac{z}{z+6}$ is purely imaginary. 2
- (b) By considering that: $\cos \theta + i \sin \theta = \cos \theta(1 + i \tan \theta)$ and de Moivre's Theorem.
- (i) Find the expression for $\cos 4\theta$ in terms of $\cos \theta$ and $\tan \theta$. 1
- (ii) As $\sin 4\theta = \cos^4 \theta (4 \tan \theta - 4 \tan^3 \theta)$, find the result for $\tan 4\theta$ in terms of $\tan \theta$. 1
- (iii) Show that the solutions to the polynomial equation: 1
- $$x^4 + 4\sqrt{3}x^3 - 6x^2 - 4\sqrt{3}x + 1 = 0$$
- can be calculated from $\tan 4\theta = \frac{1}{\sqrt{3}}$.
- (iv) Find the four solutions to this quartic equation. 2
- (v) Hence show that: $\tan \frac{7\pi}{24} \tan \frac{11\pi}{24} = \cot \frac{\pi}{24} \cot \frac{5\pi}{24}$. 2
- (c) (i) Find, in the form ' $z = cis \theta$ ', the roots of the equation: 2
- $$z^{2n+1} = 1, \text{ where } n = 0, 1, 2, \dots$$
- (ii) Hence factorise $z^{2n} + z^{2n-1} + \dots + z + 1$ into quadratic factors with real coefficients. 2
- (iii) Hence, or otherwise find 2
- $$2^n \times \sin \frac{\pi}{2n+1} \times \sin \frac{2\pi}{2n+1} \times \dots \times \sin \frac{n\pi}{2n+1}.$$

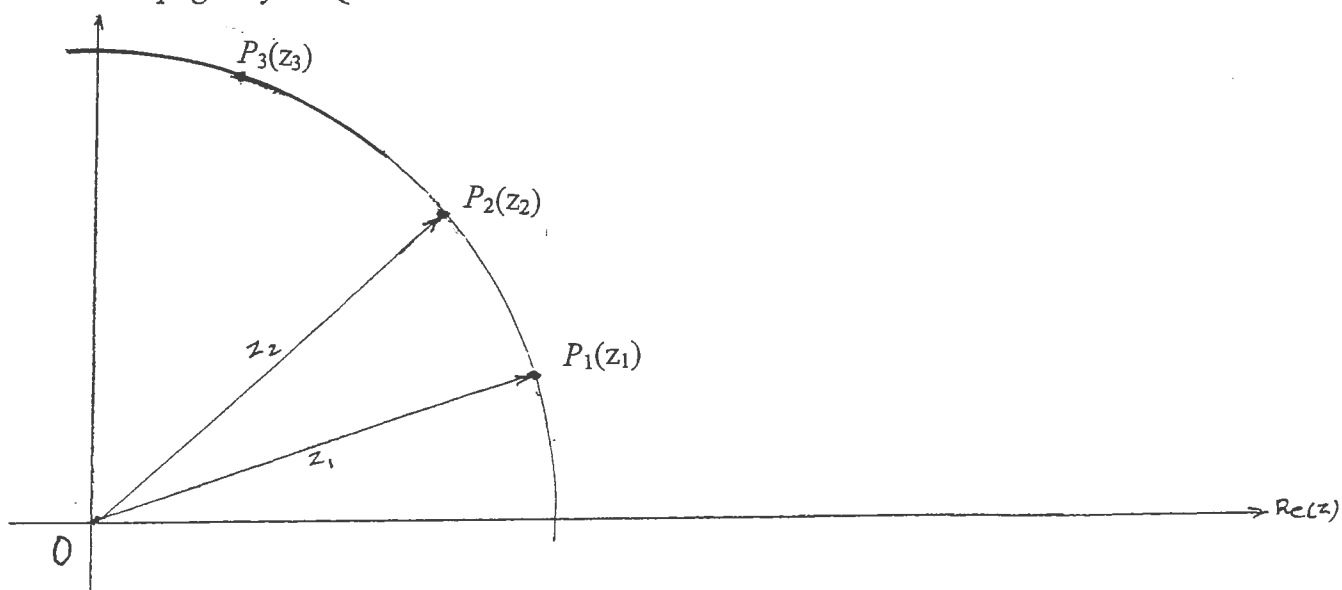
THE END



Question 3 (d)

Student id: _____

Attach this page to your Question 3 section.



MATH. EXT 2 ASSESSMENT TASK 1
TERM 4, 2011

MATHEMATICS Extension 2: Question.....		
Suggested Solutions	Marks	Marker's Comments
Q1 (a) $z_1 = 2 - 3i$ $z_2 = 3 + 4i$ (i) $z_1 = 2 - 3i = \sqrt{2^2 + (-3)^2} = \sqrt{13}$ (ii) $z_1 + \bar{z}_2 = 2 - 3i + (3 + 4i)$ $= 2 - 3i + 3 - 4i$ $= 5 - 7i$ (iii) $\frac{z_2}{z_1} = \frac{3 + 4i}{2 - 3i} \times \frac{2 + 3i}{2 + 3i} = \frac{6 + 9i + 8i - 12}{4 + 9}$ $= \frac{-6 + 17i}{13}$	1 1 12	1 11 12
(b) $(a + 3i)(2 - i) = 17 + bi$ $2a - ai + 21i + 3 = 17 + bi$ Equating Real and Imaginary parts $2a + 3 = 17 \quad (1)$ $21 - a = b \quad (2)$ (1) $2a = 14 \Rightarrow a = 2$ (2) $\therefore b = 21 - 2 = 19$	1 1	12
(c) Let $z = x + iy$; $x, y \in \mathbb{R}$ $z^2 = 35 - 12i$ $x^2 - y^2 + 2xyi = 35 - 12i$ So $x^2 - y^2 = 35 \quad (1)$ $2xy = -12 \quad (2)$ $\therefore x^2 y^2 = 36 \quad (2a)$ Subtract (1) from (2a): $x^4 - x^2 y^2 = 35x^2 \quad (1a)$ $x^4 - 36 = 35x^2$ $x^4 - 35x^2 - 36 = 0$ $(x^2 - 36)(x^2 + 1) = 0$ $\rightarrow$ no real roots $\therefore x = \pm 6$ $\therefore y = \frac{-12}{x} = \mp 2$ $\therefore z = 6 - 2i$ or $-6 + 2i$ these are various approaches		12

MATHEMATICS Extension 2: Question.....2

Suggested Solutions

Marks

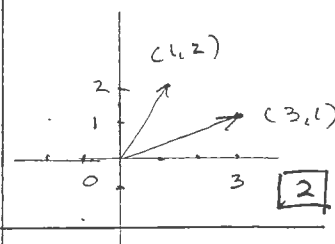
Marker's Comments

Q2 (a)

$$\begin{aligned} a+ib &= (3+i) + [1+2i - (3+i)] \times i \\ &= 3+i + [-2+i]i \\ &= 3+i - 2i - 1 \\ \therefore a+ib &= 2-i \end{aligned}$$

1

1

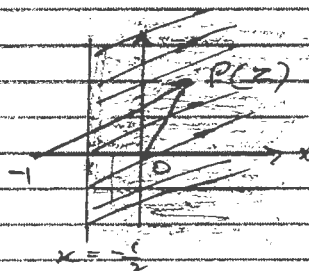


(b) $\left| \frac{1}{z} + 1 \right| \geq 1 \Rightarrow \left| \frac{1+z}{z} \right| \geq 1$

$\therefore |z+1| \geq |z|$

$(x+1)^2 + y^2 \geq x^2 + y^2$

$x \geq -\frac{1}{2}$



1

[1]

(c) (i) Since all the coefficients $(2, -19, 1, 2, a_0)$ are real
 $5-6i = 5+6i$ is also a root

$\Delta_1 = 5-6i + 5+6i + \alpha = -\frac{b}{a} = \frac{19}{2}$

$\alpha = -\frac{1}{2}$

$\therefore$ Others $5+6i$ and $-\frac{1}{2}$

1

[2]

(ii) $\Delta_2 = (5-6i)(5+6i)\left(-\frac{1}{2}\right) = -\frac{a_0}{2}$
 $= -\frac{1}{2}(25+36) = -\frac{a_0}{2}$

$\therefore a_0 = 61$

1

1

[2]

(d) Let $z = x+iy, z \neq 0$
 $\left| \frac{1}{z} + \frac{1}{\bar{z}} \right| \geq \frac{1}{2} \Rightarrow \left| \frac{z+\bar{z}}{z\bar{z}} \right| \geq \frac{1}{2}$

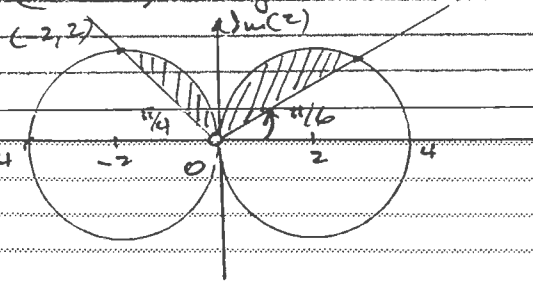
ie $\frac{|2x|}{x^2+y^2} \geq \frac{1}{2} \Rightarrow |4x| \geq x^2+y^2$

ie $x^2 - |4x| + y^2 \leq 0$

$x^2 - 4|x| + 4 + y^2 \leq 4$

$x \geq 0 \quad (x-2)^2 + y^2 \leq 4$

$x < 0 \quad (x+2)^2 + y^2 \leq 4$



shaded Region for $\frac{1}{2} \leq \operatorname{Re}(z) \leq \frac{3}{2}$

for $\operatorname{Im}(z) \geq 0$
 $\frac{3\pi}{4} \leq \operatorname{Arg} z \leq \frac{\pi}{4}$

1

$\frac{1}{2}$

and

$\left| \frac{1}{z} + \frac{1}{\bar{z}} \right| \geq \frac{1}{2}$

half Ls.doc

[4]

MATHEMATICS Extension 2: Question 2

Suggested Solutions	Marks	Marker's Comments
Q 2 (*) $z^3 = 1$ $z = 1, \omega, \bar{\omega}$ (i) $z^3 = (\omega^2)^3 = \omega^6 = (\omega^3)^2 = 1^2 = 1$ $\therefore$ by testing / using Factor theorem $z = \omega^2$ is also a root. (ii) $1 + \omega + \omega^2 = 0$ (from factor $(1+z+z^2)$) (iii) $(\omega - 1)(1 + 2\omega + 3\omega^2)$ $= (\omega - 1)(1 + \omega + \omega^2 + \omega + 2\omega^2)$ $= (\omega - 1)(\omega + 2\omega^2)$ $= \omega^2 + 2\omega^3 - \omega - 2\omega^2$ $= -\omega^2 - \omega + 2$ $= -(\omega + \omega^2) + 2$ $= -(-1) + 2$ $= 3$ many thanks to "3"	1 1 1	[1] [1] [2]

MATHEMATICS Extension 2: Question.....3

Suggested Solutions

Marks

Marker's Comments

Q3(a)(i) $z = \cos \theta + i \sin \theta$

$$z^n = (\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$$

$$z^{-n} = (\cos \theta + i \sin \theta)^{-n} = \cos(-n\theta) + i \sin(-n\theta)$$

$$= \cos n\theta - i \sin n\theta$$

$$z + z^{-n} = 2 \cos n\theta$$

$$\therefore \frac{z^n + z^{-n}}{2} = \cos n\theta \quad \text{q.e.d.}$$

$\frac{1}{2}$ (De Moivre's thm)

(" ")

$\frac{1}{2}$

$\frac{1}{2}$

[2]

(ii) $\cos \theta \cdot \cos 3\theta = \frac{1}{2}(z + z^{-1}) \times \frac{1}{2}(z^3 + z^{-3})$

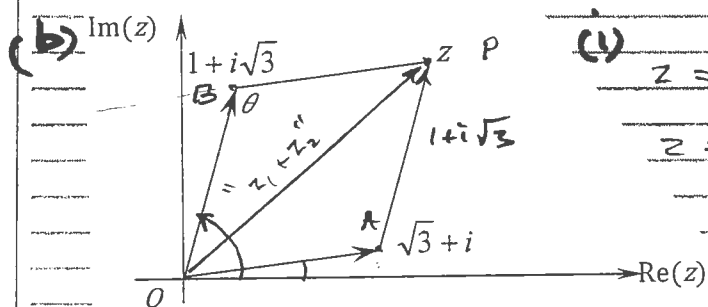
$$= \frac{1}{4} [z^3 + z^{-1} + z^{-3} + z]$$

$$= \frac{1}{4} [z + z^{-1} + z^3 + z^{-3}]$$

$$= \frac{1}{4} [2 \cos \theta + 2 \cos 3\theta]$$

$$= \frac{1}{2} [\cos \theta + \cos 3\theta]$$

[2]



(i) $z = \sqrt{3} + i + 1 + i\sqrt{3}$

$$z = (1 + \sqrt{3}) + i(1 + \sqrt{3})$$

$$= (1 + \sqrt{3})(1 + i)$$

(Vector addition)

[1]

(ii) $\text{Arg}(\sqrt{3} + i) = \frac{\pi}{6}$

$$\text{Arg}(1 + i\sqrt{3}) = \frac{\pi}{3}$$

$$\therefore \angle AOB = \frac{\pi}{3} - \frac{\pi}{6} = \frac{\pi}{6}$$

$$\therefore \theta = \pi - \frac{\pi}{6} = \frac{5\pi}{6} \quad (\text{co-interior angles are supplementary as } BP \parallel OA)$$

Many ways to arrive at $\frac{5\pi}{6}$

1

[2]

(c) NOTE ORDER

$$z^5 + z^2 - z + k = 0, k \in \mathbb{R}$$

Degree 5 all coefficients real

$\therefore$ could have 5 real roots, 3 + 1 conjugate pair, or 1 real + 2 pair of conjugate roots

Diagram B

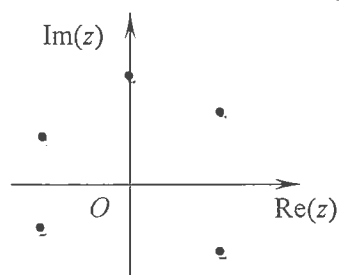


Diagram A

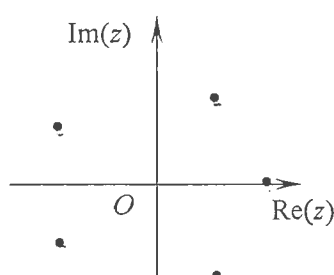


Diagram B

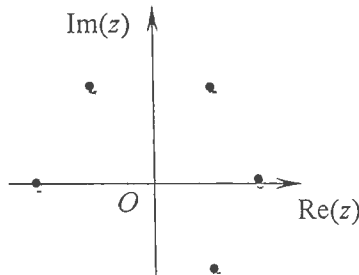


Diagram C

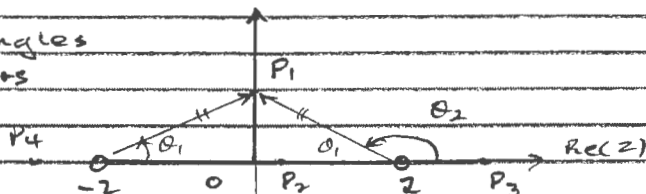
[2]

5.

Q3(c) $\text{Arg}(z+2) + \text{Arg}(z-2) = \pi$

* Locus for z : $-2 < x < 2$ or $y \neq 0$ *

• Using Angles and Vectors



$\text{Arg}(z+2) = \theta_1$

CHECK/test $P_1(0, ki)$

$\text{Arg}(z-2) = \theta_2$

$\checkmark \checkmark P_{2,3,4}(x, 0)$

• $\tan(\theta_1 + \theta_2) = 0$

$\tan \theta_1 + \tan \theta_2 = 0$

$1 - \tan \theta_1 \tan \theta_2$

$\Rightarrow \tan \theta_1 + \tan \theta_2 = 0$ BUT $\tan \theta_1 \tan \theta_2 \neq 1$

i.e. $\frac{y}{x+2} + \frac{y}{x-2} = 0$ | $\frac{y^2}{x^2-4} \neq 1$

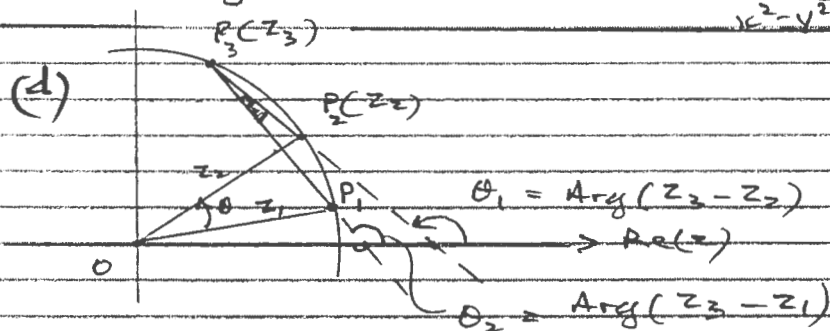
$\frac{2xy}{x^2-4} = 0$ | $\Rightarrow x^2 - y^2 \neq 4$

$\therefore x=0$ or $y=0$ but $x \neq \pm 2$

Need to test as incomplete information from this algebra

• $\text{Arg}[(z+2)(z-2)] = \pi$ if

$\tan \text{Arg}(z^2-4) = 0 \Rightarrow \frac{2xy}{x^2-y^2-4} = 0$



Let $\theta_1 = \text{Arg}(z_3 - z_2)$

$\theta_2 = \text{Arg}(z_3 - z_1)$

$\theta = \text{Arg}(z_2) - \text{Arg}(z_1) = \text{Arg}\left(\frac{z_2}{z_1}\right)$

$\theta_1 = \theta_2 + \angle P_2 P_3 P_1$ (Exterior angle of triangle equals sum of interior opposite angles)

$\therefore \angle P_2 P_3 P_1 = \theta_1 - \theta_2 = \text{Arg}(z_3 - z_2) - \text{Arg}(z_3 - z_1) = \text{Arg}\left(\frac{z_3 - z_2}{z_3 - z_1}\right)$

BUT $\angle P_1 O P_2 = \theta = 2 \angle P_2 P_3 P_1$ (Angle subtended at centre is twice the angle subtended at circumference).

$\therefore \angle P_2 P_3 P_1 = \text{Arg}\left(\frac{z_3 - z_2}{z_3 - z_1}\right) = \frac{1}{2} \angle P_1 O P_2 = \frac{1}{2} \text{Arg}\left(\frac{z_2}{z_1}\right)$

MATHEMATICS Extension 2: Question.....4

Suggested Solutions

Marks

Marker's Comments

Q4(a)

Let $z = x + iy$

If z is purely imaginary

$\therefore \operatorname{Re}\left(\frac{z}{z+b}\right) = 0$ or $\operatorname{Arg}\left(\frac{z}{z+b}\right) = \pm \frac{\pi}{2}$

$\frac{z}{z+b} = \frac{x+iy}{x+b+iy} \times \frac{(x+b)-iy}{(x+b)-iy}$

$= \frac{x^2+bx+y^2+i(xy+by-xy)}{(x+b)^2+y^2} = \frac{x^2+bx+y^2+byi}{(x+b)^2+y^2}$

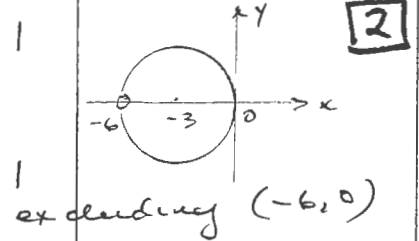
$\therefore \operatorname{Re}\left(\frac{z}{z+b}\right) = \frac{x^2+bx+y^2}{(x+b)^2+y^2} = 0$

$\Rightarrow x^2+bx+y^2 = 0$ with $z \neq -b$

i.e. $(x+3)^2+y^2 = 9$

the locus is a circle $C(-3, 0)$ radius 3

as $\operatorname{Im}\left(\frac{z}{z+b}\right) = ki$



excluding $(-6, 0)$

(i) $\cos \theta + i \sin \theta = \cos \theta (1 + i \tan \theta)$

(b) $\therefore (\cos \theta + i \sin \theta)^4 = \cos^4 \theta (1 + i \tan \theta)^4$

$\cos 4\theta + i \sin 4\theta = \cos^4 \theta (1 + 4it + 6i^2t^2 + 4i^3t^3 + i^4t^4)$
 $= \cos^4 \theta (1 - 6\tan^2 \theta + \tan^4 \theta + i(4\tan \theta - 4\tan^3 \theta))$

equating real part

$\therefore \cos 4\theta = \cos^4 \theta (1 - 6\tan^2 \theta + \tan^4 \theta)$

(ii) $\tan 4\theta = \frac{\sin 4\theta}{\cos 4\theta} = \frac{\cos^4 \theta (4\tan \theta - 4\tan^3 \theta)}{\cos^4 \theta (1 - 6\tan^2 \theta + \tan^4 \theta)}$
 $= \frac{4\tan \theta - 4\tan^3 \theta}{1 - 6\tan^2 \theta + \tan^4 \theta}$

(iii) $x^4 + 4\sqrt{3}x^3 - 6x^2 - 4\sqrt{3}x + 1 = 0$
 $\sqrt{3}(4x - 4x^3) = x^4 - 6x^2 + 1$

so $\frac{4x - 4x^3}{x^4 - 6x^2 + 1} = \frac{1}{\sqrt{3}}$

By letting $x = \tan \theta$

so $\frac{4\tan \theta - 4\tan^3 \theta}{1 - 6\tan^2 \theta + \tan^4 \theta} = \tan 4\theta = \frac{1}{\sqrt{3}}$

$\therefore$ solutions can be found for the equation from $\tan 4\theta = \frac{1}{\sqrt{3}}$

(iv) $\tan 4\theta = \frac{1}{\sqrt{3}} \therefore 4\theta = n\pi + \frac{\pi}{6} \quad n \in \mathbb{Z}$
 $\theta = (6n+1)\frac{\pi}{24}$

$\therefore \theta_1 = \frac{\pi}{24}, \theta_2 = \frac{7\pi}{24}, \theta_3 = \frac{13\pi}{24}, \theta_4 = \frac{19\pi}{24}$

$\therefore x_1 = \tan \frac{\pi}{24}, x_2 = \tan \frac{7\pi}{24}, x_3 = \tan \frac{13\pi}{24}, x_4 = \tan \frac{19\pi}{24}$
 $= -\tan \frac{11\pi}{24}, = -\tan \frac{5\pi}{24}$

(v) $\Delta_4 = x_1 x_2 x_3 x_4 = \frac{e}{a}$

$\tan \frac{\pi}{24} \times \tan \frac{7\pi}{24} \times (-\tan \frac{11\pi}{24}) \times (-\tan \frac{5\pi}{24}) = 1$

$\therefore \tan \frac{\pi}{24} \cdot \tan \frac{7\pi}{24} = +\cot \frac{5\pi}{24} \cdot \cot \frac{11\pi}{24}$

4. MATHEMATICS Extension 2: Question.....

Suggested Solutions	Marks	Marker's Comments
Q4 (c) (i) $z^{2n+1} = 1$; $z = r \text{cis} \theta$ $r^{2n+1} (\text{cis} \theta)^{2n+1} = 1 = 1 \cdot \text{cis} (2k\pi)$ $k \in \mathbb{Z}$ $r^{2n+1} \text{cis} ((2n+1)\theta) = 1 \text{cis} (2k\pi)$ $\therefore r = 1$ $(2n+1)\theta = 2k\pi$ $\theta = \frac{2k\pi}{2n+1}$ ✓ $\therefore$ roots are going to be $z_k = 1 \text{cis} \left(\frac{2k\pi}{2n+1} \right)$ $k = 0, \pm 1, \pm 2, \dots, n$ ie $\text{cis} 0, \text{cis} \frac{2\pi}{2n+1}, \text{cis} \frac{4\pi}{2n+1}, \text{cis} \frac{6\pi}{2n+1}, \dots$		solutions separated by $\theta = \frac{2\pi k}{2n+1}$ 2
(ii) $z^{2n+1} - 1 = (z-1)[z^{2n} + z^{2n-1} + \dots + z + 1]$ $\therefore z = 1 = \text{cis} 0$ is a root (for $k=0$). All other roots come in conjugate pairs α and $\bar{\alpha}$ since all coeffs real. So $(z - \alpha_k)(z - \bar{\alpha}_k) = z^2 - 2\text{Re}(\alpha_k)z + \alpha_k ^2$ $= z^2 - 2\cos \theta_k z + 1$, where $\alpha_k = \text{cis} \frac{2k\pi}{2n+1}$ So $z^{2n} + z^{2n-1} + \dots + z + 1$ $= \left[z^2 - 2\cos \frac{2\pi}{2n+1} z + 1 \right] \left[z^2 - 2\cos \frac{4\pi}{2n+1} z + 1 \right] \dots \left[z^2 - 2\cos \frac{2n\pi}{2n+1} z + 1 \right]$ factorised over the reals. $\sum_{m=0}^{2n} z^m = \prod_{k=1}^n \left[z^2 - 2\cos \frac{2k\pi}{2n+1} z + 1 \right]$		2
(iii) Let $z = 1$ in (ii) $1 + 1 + \dots + 1 + 1 = \left[2 - 2\cos \frac{2\pi}{2n+1} \right] \left[2 - 2\cos \frac{4\pi}{2n+1} \right] \dots \left[2 - 2\cos \frac{2n\pi}{2n+1} \right]$ $(2n+1) \times 1 = 2^n \left(1 - \cos \frac{2\pi}{2n+1} \right) \left(1 - \cos \frac{4\pi}{2n+1} \right) \dots \left(1 - \cos \frac{2n\pi}{2n+1} \right)$ n-pairs $2n+1 = 2^n \times 2\sin^2 \frac{\pi}{2n+1} \times 2\sin^2 \frac{2\pi}{2n+1} \times \dots \times 2\sin^2 \frac{n\pi}{2n+1}$ $= 2^n \times 2^n \times \sin^2 \frac{\pi}{2n+1} \times \sin^2 \frac{2\pi}{2n+1} \times \dots \times \sin^2 \frac{n\pi}{2n+1}$ $2n+1 = \left(2^n \sin \frac{\pi}{2n+1} \times \sin \frac{2\pi}{2n+1} \times \dots \times \sin \frac{n\pi}{2n+1} \right)^2$ $\therefore 2^n \sin \frac{\pi}{2n+1} \times \sin \frac{2\pi}{2n+1} \times \dots \times \sin \frac{n\pi}{2n+1} = \sqrt{2n+1}$		2